

Quiver Hecke algebra (KLR-algebra)

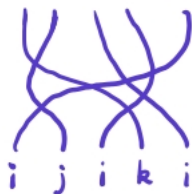
Let I be a set, k be a field, $Q = (Q_{ij})_{i,j \in I}$ be a matrix in $k^{I \times I}$ with $Q_{ii} = 0$ for $i \in I$. Fix $n \in \mathbb{N}$ and $\alpha \in \mathbb{N}^I$ such that $\text{ht}(\alpha) = n$. We define a subset

$$I(\alpha) = \{ \nu \in I^n \mid |\nu| = \alpha \}$$

where $|\nu|$ is the type of ν . We define a k -algebra $R(\alpha)$: it has generators 1_ν , $\nu \in I(\alpha)$, $x_1, \dots, x_n$ and $\tau_1, \dots, \tau_{n-1}$, subject to the following relations:

- $1_\nu 1_{\nu'} = \delta_{\nu, \nu'} 1_\nu$
- $\tau_i 1_\nu = 1_{s_i(\nu)} \tau_i$
- $x_i 1_\nu = 1_\nu x_i$
- $x_i x_j 1_\nu = x_j x_i 1_\nu$
- $\tau_i^2 1_\nu = Q_{\nu_i, \nu_{i+1}}(x_i 1_\nu, x_{i+1} 1_\nu)$
- $\tau_i \tau_j 1_\nu = \tau_j \tau_i 1_\nu$ if $|i-j| > 1$.
- $\tau_{i+1} \tau_i \tau_{i+1} 1_\nu - \tau_i \tau_{i+1} \tau_i 1_\nu = \delta_{\nu_i, \nu_{i+2}} \frac{Q_{\nu_i, \nu_{i+1}}(x_i 1_\nu, x_{i+1} 1_\nu) - Q_{\nu_i, \nu_{i+1}}(x_{i+2} 1_\nu, x_{i+1} 1_\nu)}{x_i 1_\nu - x_{i+2} 1_\nu}$
- $\tau_i x_j 1_\nu - x_{s_i(j)} \tau_i 1_\nu = \begin{cases} (\delta_{i+1, j} - \delta_{i, j}) 1_\nu & \text{if } \nu_i = \nu_{i+1} \\ 0 & \text{otherwise} \end{cases}$

The algebra $R(\alpha)$ has the following diagrammatic description: elements of $R(\alpha)$ are diagrams of colored strands with dots:



(in other words, each strand has a label by element of I). We allow k -linear combinations of such diagrams, modulo the relations:

$$(1) \quad \text{Diagram of two strands } i, j \text{ crossing} = \text{Diagram of two parallel strands } i, j \text{ with a box labeled } Q_{ij}(u, v)$$

$$(2) \quad \text{Diagram of strands } i, j \text{ crossing with dot on } i = \delta_{ij} \text{ Diagram of two parallel strands } i, j$$

$$\text{Diagram of strands } i, j \text{ crossing with dot on } j = \delta_{ij} \text{ Diagram of two parallel strands } i, j$$

$$(3) \quad \begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} - \begin{array}{c} \text{Diagram 3} \\ \text{Diagram 4} \end{array} = \delta_{i,k} \begin{array}{c} \text{Diagram 5} \\ \text{Diagram 6} \end{array} \boxed{H_{ij}(u,v,w)}$$

$$\text{where } H_{ij}(u,v,w) = \frac{Q_{ij}(u,v) - Q_{ij}(w,v)}{u-w}$$

Let $\mathcal{S}$ be a set of finite sequences of elements in $\{1, 2, \dots, n+1\}$ such that $\{S_1 \dots S_k\}_{(i_1, \dots, i_k) \in J}$ is a set of reduced expressions of elements of S_n . Then

$\mathcal{B}_\alpha = \{x_1^{a_1} \dots x_n^{a_n} \cdot \tau_{i_1} \dots \tau_{i_k} 1_v \mid v \in I(\alpha), (i_1, \dots, i_k) \in \mathcal{S}, (a_1, \dots, a_n) \in \mathbb{N}^n\}$
generate $R(\alpha)$ as a k -module.

Lemma. If $\mathcal{B}_\alpha$ is linearly independent, then $Q_{ij}(u,v) = Q_{ji}(v,u)$ for $i, j \in I$.

Proof. Let $v \in I(\alpha)$ with $v_i \neq v_{i+1}$. Then

$$\tau_i \tau_i \tau_i 1_v = \tau_i \tau_i 1_{s_i(v)} \tau_i 1_v = Q_{v_{i+1}, v_i}(x_i 1_{s_i(v)}, x_{i+1} 1_{s_i(v)}) \tau_i 1_v$$

$$\tau_i \tau_i \tau_i 1_v = \tau_i 1_v \tau_i \tau_i 1_v = \tau_i 1_v Q_{v_i, v_{i+1}}(x_i 1_v, x_{i+1} 1_v) = Q_{v_i, v_{i+1}}(x_{i+1} 1_{s_i(v)}, x_i 1_{s_i(v)}) \tau_i 1_v$$

$$\Rightarrow \left(Q_{v_{i+1}, v_i}(x_i 1_{s_i(v)}, x_{i+1} 1_{s_i(v)}) - Q_{v_i, v_{i+1}}(x_{i+1} 1_{s_i(v)}, x_i 1_{s_i(v)}) \right) \tau_i 1_v = 0$$

so we must have $Q_{v_{i+1}, v_i}(x_i 1_{s_i(v)}, x_{i+1} 1_{s_i(v)}) = Q_{v_i, v_{i+1}}(x_{i+1} 1_{s_i(v)}, x_i 1_{s_i(v)})$, i.e. $Q_{ij}(u,v) = Q_{ji}(v,u)$.

In the following we always assume $Q_{ij}(u,v) = Q_{ji}(v,u)$. Such matrix Q will be called hermitian.

To show that $\mathcal{B}_\alpha$ is a basis of $R(\alpha)$, we define a polynomial representation of $R(\alpha)$.

Let $P = (P_{ij})_{i,j \in I}$ be a matrix in $k[u,v]$ with $P_{ii} = 0$. Assume that we have $Q_{ij}(u,v) = P_{ij}(u,v) P_{ji}(v,u)$ (for example, we can take $P_{ij} = \begin{cases} Q_{ij} & i > j \\ 0 & i \leq j \end{cases}$, where $<$ is a total order on I).

Then we define a representation of $R(\alpha)$ on $\text{Pol}_\alpha = \bigoplus_{v \in I(\alpha)} \text{Pol}_v$, where $\text{Pol}_v = k[x_{1,v}, \dots, x_{n,v}]$, by

$$1_v(f) := \delta_{v,v'} f \quad \forall f \in \text{Pol}_{v'}$$

$$x_i 1_v(f) := \delta_{v,v'} x_{i,v} f \quad \forall f \in \text{Pol}_{v'}$$

$$\tau_i 1_v(f) := \begin{cases} \frac{s_i(f) - f}{x_{i,v} - x_{i+1,v}} & \text{if } f \in \text{Pol}_v, v_i = v_{i+1} \\ P_{v_i, v_{i+1}}(x_{i+1, s_i(v)}, x_{i, s_i(v)}) s_i(f) & \text{if } f \in \text{Pol}_v, v_i \neq v_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

Here the symmetric group S_n acts on Pol_α by sending $x_{i,v}$ to $x_{w(i),w(v)}$.

Lemma. These give a well-defined representation of $R(\alpha)$.

Proof. The first four relations are immediate. Now for $f \in \text{Pol}_\nu$, we have

$$\begin{aligned}
 \tau_i^2 \mathbb{I}_\nu(f) &= \tau_i \mathbb{I}_{s_i(\nu)} \tau_i \mathbb{I}_\nu(f) \\
 &= \begin{cases} \tau_i \mathbb{I}_{s_i(\nu)} \partial_i(f) & \text{if } \nu_i = \nu_{i+1} \\ \tau_i \mathbb{I}_{s_i(\nu)} (P_{\nu_i, \nu_{i+1}}(x_{i+1}, x_i) s_i(f)) & \text{if } \nu_i \neq \nu_{i+1} \end{cases} \\
 \partial_i &= \text{Demazure operator} \\
 &= \begin{cases} \partial_i^2 f & \text{if } \nu_i = \nu_{i+1} \\ P_{\nu_{i+1}, \nu_i}(x_i, x_{i+1}) s_i(P_{\nu_i, \nu_{i+1}}(x_{i+1}, x_i) s_i(f)) & \text{if } \nu_i \neq \nu_{i+1} \end{cases} \\
 &= \begin{cases} 0 & \text{if } \nu_i = \nu_{i+1} \\ P_{\nu_{i+1}, \nu_i}(x_i, x_{i+1}) P_{\nu_i, \nu_{i+1}}(x_i, x_{i+1}) f & \text{if } \nu_i \neq \nu_{i+1} \end{cases} \\
 &= \begin{cases} 0 & \text{if } \nu_i = \nu_{i+1} \\ Q_{\nu_i, \nu_{i+1}}(x_i, x_{i+1}) f & \text{if } \nu_i \neq \nu_{i+1} \end{cases} \\
 &= Q_{\nu_i, \nu_{i+1}}(x_i \mathbb{I}_\nu, x_{i+1} \mathbb{I}_\nu)(f).
 \end{aligned}$$

It is also clear that $\tau_i \tau_j \mathbb{I}_\nu(f) = \tau_j \tau_i \mathbb{I}_\nu(f)$ if $|i-j| > 1$.

Now for $f \in \text{Pol}_\nu$,

$$\alpha_{i+1, \nu} = x_{i+1, \nu} - x_{i+2, \nu}$$

$$\begin{aligned}
 \tau_i \tau_{i+1} \mathbb{I}_\nu(f) &= \begin{cases} \tau_i \mathbb{I}_{s_{i+1}(\nu)} \left(\frac{s_{i+1}(f) - f}{\alpha_{i+1}} \right) & \text{if } \nu_{i+1} = \nu_{i+2} \\ \tau_i \mathbb{I}_{s_{i+1}(\nu)} (P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_{i+1}) s_{i+1}(f)) & \text{if } \nu_{i+1} \neq \nu_{i+2} \end{cases} \\
 \begin{array}{ccc} \nu_i & \nu_{i+1} & \nu_{i+2} \\ & \downarrow s_{i+1} & \\ \nu_i & \nu_{i+2} & \nu_{i+1} \\ & \downarrow s_i & \\ \nu_{i+2} & \nu_i & \nu_{i+1} \\ & \downarrow s_{i+1} & \\ \nu_{i+2} & \nu_{i+1} & \nu_i \end{array} &= \begin{cases} \frac{\alpha_{i+1} s_i s_{i+1}(f) - \alpha_{i+1} s_i(f) - (\alpha_i + \alpha_{i+1}) s_{i+1}(f) + (\alpha_i + \alpha_{i+1}) f}{\alpha_i \alpha_{i+1} (\alpha_i + \alpha_{i+1})} & \text{if } \nu_i = \nu_{i+2}, \nu_{i+1} = \nu_{i+2} \\ P_{\nu_i, \nu_{i+2}}(x_{i+1}, x_i) \cdot \frac{s_i s_{i+1}(f) - s_i(f)}{\alpha_i + \alpha_{i+1}} & \text{if } \nu_i \neq \nu_{i+2}, \nu_{i+1} = \nu_{i+2} \\ \frac{P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_i) s_i s_{i+1}(f) - P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_{i+1}) s_{i+1}(f)}{\alpha_i} & \text{if } \nu_i = \nu_{i+2}, \nu_{i+1} \neq \nu_{i+2} \\ P_{\nu_i, \nu_{i+2}}(x_{i+1}, x_i) P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_i) s_i s_{i+1}(f) & \text{if } \nu_i \neq \nu_{i+2}, \nu_{i+1} \neq \nu_{i+2} \end{cases}
 \end{aligned}$$

Similarly,

$$\tau_{i+1} \tau_i \tau_i(f) = \begin{cases} \tau_{i+1} \tau_{i+1} \left(\frac{s_i(f) - f}{\alpha_i} \right) & \text{if } v_i = v_{i+1} \\ \tau_{i+1} \tau_{i+1} (P_{v_i, v_{i+1}}(x_{i+1}, x_i) s_i(f)) & \text{if } v_i \neq v_{i+1} \end{cases}$$

$$\begin{array}{ccc} v_i & v_{i+1} & v_{i+2} \\ \downarrow s_i & & \\ v_{i+1} & v_i & v_{i+2} \\ \downarrow s_{i+1} & & \\ v_{i+1} & v_{i+2} & v_i \\ \downarrow s_i & & \\ v_{i+2} & v_{i+1} & v_i \end{array} = \begin{cases} \frac{\alpha_i s_{i+1} s_i(f) - \alpha_i s_{i+1}(f) - (\alpha_i + \alpha_{i+1}) s_i(f) + (\alpha_i + \alpha_{i+1}) f}{\alpha_i \alpha_{i+1} (\alpha_i + \alpha_{i+1})} & \text{if } v_i = v_{i+1}, v_i = v_{i+2} \\ P_{v_i, v_{i+2}}(x_{i+2}, x_{i+1}) \cdot \frac{s_{i+1} s_i(f) - s_{i+1}(f)}{\alpha_i + \alpha_{i+1}} & \text{if } v_i = v_{i+1}, v_i \neq v_{i+2} \\ \frac{P_{v_i, v_{i+1}}(x_{i+2}, x_i) s_{i+1}(f) - P_{v_i, v_{i+1}}(x_{i+1}, x_i) s_i(f)}{\alpha_{i+1}} & \text{if } v_i \neq v_{i+1}, v_i = v_{i+2} \\ P_{v_i, v_{i+2}}(x_{i+2}, x_{i+1}) P_{v_i, v_{i+1}}(x_{i+2}, x_i) s_{i+1} s_i(f) & \text{if } v_i \neq v_{i+1}, v_i \neq v_{i+2} \end{cases}$$

Case a. $v_i = v_{i+1} = v_{i+2}$. Then

$$\tau_{i+1} \tau_i \tau_i(f) = \partial_{i+1} \partial_i \partial_{i+1}(f) = \tau_i \tau_{i+1} \tau_i(f)$$

Case b. $v_i \neq v_{i+1}, v_{i+1} = v_{i+2}$. Then

$$\begin{aligned} \tau_{i+1} \tau_i \tau_i(f) &= \tau_{i+1} \left(P_{v_i, v_{i+2}}(x_{i+1}, x_i) \cdot \frac{s_i s_{i+1}(f) - s_i(f)}{\alpha_i + \alpha_{i+1}} \right) \\ &= P_{v_{i+1}, v_i}(x_{i+2}, x_{i+1}) s_{i+1} \left(P_{v_i, v_{i+1}}(x_{i+1}, x_i) \cdot \frac{s_i s_{i+1}(f) - s_i(f)}{\alpha_i + \alpha_{i+1}} \right) \\ &= P_{v_{i+1}, v_i}(x_{i+2}, x_{i+1}) P_{v_i, v_{i+1}}(x_{i+2}, x_i) \frac{s_{i+1} s_i s_{i+1}(f) - s_{i+1} s_i(f)}{\alpha_i} \end{aligned}$$

Similarly, $\tau_i \tau_{i+1} \tau_i(f) = \tau_i \tau_{i+1} (P_{v_i, v_{i+1}}(x_{i+1}, x_i) s_i(f))$

$$\begin{aligned} &= \tau_i (P_{v_i, v_{i+2}}(x_{i+2}, x_{i+1}) s_{i+1} (P_{v_i, v_{i+1}}(x_{i+1}, x_i) s_i(f))) \\ &= \tau_i (P_{v_i, v_{i+2}}(x_{i+2}, x_{i+1}) P_{v_i, v_{i+1}}(x_{i+2}, x_i) s_{i+1} s_i(f)) \\ &= \frac{P_{v_i, v_{i+2}}(x_{i+2}, x_i) P_{v_i, v_{i+1}}(x_{i+2}, x_{i+1}) s_i s_{i+1} s_i(f) - P_{v_i, v_{i+2}}(x_{i+2}, x_{i+1}) P_{v_i, v_{i+1}}(x_{i+2}, x_i) s_{i+1} s_i(f)}{\alpha_i} \end{aligned}$$

Since $v_{i+1} = v_{i+2}$, we see that $\tau_{i+1} \tau_i \tau_i(f) = \tau_i \tau_{i+1} \tau_i(f)$. The case $v_i = v_{i+1}, v_{i+1} \neq v_{i+2}$ is similar.

Case c. v_i, v_{i+1}, v_{i+2} are distinct. Then

$$\begin{aligned} \tau_{i+1} \tau_i \tau_i(f) &= P_{v_i, v_{i+1}}(x_{i+2}, x_{i+1}) s_{i+1} (P_{v_i, v_{i+2}}(x_{i+1}, x_i) P_{v_{i+1}, v_{i+2}}(x_{i+2}, x_i) s_i s_{i+1}(f)) \\ &= P_{v_i, v_{i+1}}(x_{i+2}, x_{i+1}) P_{v_i, v_{i+2}}(x_{i+2}, x_i) P_{v_{i+1}, v_{i+2}}(x_{i+1}, x_i) s_{i+1} s_i s_{i+1}(f) \\ \tau_i \tau_{i+1} \tau_i(f) &= P_{v_{i+1}, v_{i+2}}(x_{i+1}, x_i) s_i (P_{v_i, v_{i+2}}(x_{i+2}, x_{i+1}) s_{i+1} (P_{v_i, v_{i+1}}(x_{i+1}, x_i) s_i(f))) \\ &= P_{v_{i+1}, v_{i+2}}(x_{i+1}, x_i) P_{v_i, v_{i+2}}(x_{i+2}, x_i) P_{v_i, v_{i+1}}(x_{i+2}, x_{i+1}) s_i s_{i+1} s_i(f) \end{aligned}$$

so $\tau_{i+1} \tau_i \tau_i(f) = \tau_i \tau_{i+1} \tau_i(f)$.

Case d. $\nu_i \neq \nu_{i+1}$, $\nu_{i+1} \neq \nu_{i+2}$, but $\nu_i = \nu_{i+2}$. Then

$$\begin{aligned}\tau_{i+1} \tau_i \tau_{i+1}(f) &= \tau_{i+1} \left(\frac{P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_i) S_i S_{i+1}(f) - P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_{i+1}) S_{i+1}(f)}{\alpha_i} \right) \\ &= P_{\nu_i, \nu_{i+1}}(x_{i+2}, x_{i+1}) S_{i+1} \left(\frac{P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_i) S_i S_{i+1}(f) - P_{\nu_{i+1}, \nu_{i+2}}(x_{i+2}, x_{i+1}) S_{i+1}(f)}{\alpha_i} \right) \\ &= \frac{P_{\nu_i, \nu_{i+1}}(x_{i+2}, x_{i+1}) P_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_i) S_{i+1} S_i S_{i+1}(f) - P_{\nu_i, \nu_{i+1}}(x_{i+2}, x_{i+1}) P_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_{i+2}) f}{\alpha_i + \alpha_{i+1}}\end{aligned}$$

$$\begin{aligned}\tau_i \tau_{i+1} \tau_i(f) &= \tau_i \left(\frac{P_{\nu_i, \nu_{i+1}}(x_{i+2}, x_i) S_{i+1} S_i(f) - P_{\nu_i, \nu_{i+1}}(x_{i+1}, x_i) S_i(f)}{\alpha_{i+1}} \right) \\ &= P_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_i) S_i \left(\frac{P_{\nu_i, \nu_{i+1}}(x_{i+2}, x_i) S_{i+1} S_i(f) - P_{\nu_i, \nu_{i+1}}(x_{i+1}, x_i) S_i(f)}{\alpha_{i+1}} \right) \\ &= \frac{P_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_i) P_{\nu_i, \nu_{i+1}}(x_{i+2}, x_{i+1}) S_i S_{i+1} S_i(f) - P_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_i) P_{\nu_i, \nu_{i+1}}(x_i, x_{i+1}) f}{\alpha_i + \alpha_{i+1}}\end{aligned}$$

$$\text{So } \tau_{i+1} \tau_i \tau_{i+1}(f) - \tau_i \tau_{i+1} \tau_i(f) = \frac{P_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_i) P_{\nu_i, \nu_{i+1}}(x_i, x_{i+1}) f - P_{\nu_i, \nu_{i+1}}(x_{i+2}, x_{i+1}) P_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_{i+2}) f}{\alpha_i + \alpha_{i+1}}$$

$$\begin{aligned}(\nu_i = \nu_{i+2}) &= \frac{Q_{\nu_i, \nu_{i+1}}(x_i, x_{i+1}) - Q_{\nu_{i+1}, \nu_{i+2}}(x_{i+1}, x_{i+2})}{x_i - x_{i+2}} (f) \\ &= \frac{Q_{\nu_i, \nu_{i+1}}(x_i, x_{i+1}) - Q_{\nu_i, \nu_{i+1}}(x_{i+2}, x_{i+1})}{x_i - x_{i+2}} (f)\end{aligned}$$

We have therefore verified the braid relation.

Finally, for $f \in \text{Pol}_\nu$, we have

$$\tau_i x_j(f) = \tau_i(x_j f) = \begin{cases} \frac{S_i(x_j f) - x_j f}{\alpha_i} & \text{if } \nu_i = \nu_{i+1} \\ P_{\nu_i, \nu_{i+1}}(x_{i+1}, x_i) S_i(x_j f) & \text{if } \nu_i \neq \nu_{i+1} \end{cases}$$

and

$$x_{S(i,j)} \tau_i(f) = \begin{cases} x_{S(i,j)} \frac{S_i(f) - f}{\alpha_i} & \text{if } \nu_i = \nu_{i+1} \\ x_{S(i,j)} P_{\nu_i, \nu_{i+1}}(x_{i+1}, x_i) S_i(f) & \text{if } \nu_i \neq \nu_{i+1} \end{cases}$$

Thus $\tau_i x_j(f) = x_{S(i,j)} \tau_i(f)$ if $\nu_i \neq \nu_{i+1}$. If $\nu_i = \nu_{i+1}$, this reduces to the relation for NH_n .

Prop. Assume that for each i, j , $P_{ij} \neq 0$. Then the representation of $R(\omega)$ on Pol_ω is faithful, and B_ω is a k -basis of $R(\omega)$.

Proof. Consider $1_\nu R(\omega) 1_\nu$. We prove that

$$\nu B_\nu := \{ x_1^{a_1} \cdots x_n^{a_n} \cdot \tau_{i_1} \cdots \tau_{i_r} \mid (a_1, \dots, a_n) \in \mathbb{N}^n, (i_1, \dots, i_r) \in I, s_{i_1} \cdots s_{i_r}(\omega) = \nu \}$$

is a k -basis for $1_\nu R(\omega) 1_\nu$.

Note that we can write

$$\tau_i 1_\nu \in \text{Hom}(P_\nu, P_{s_i(\nu)}) \cong \text{End}(k[x_1, \dots, x_n])$$

then
$$\tau_i 1_\nu = \begin{cases} \partial_i \\ p_i(x_1, \dots, x_n) \partial_i + p_i \end{cases} = Q_i \partial_i + P_i$$

And in general, for $w \in \nu S_\nu$, if we think

$$\tau_{i_1} \cdots \tau_{i_r} 1_\nu \in \text{Hom}(P_\nu, P_\nu) \cong \text{End}(k[x_1, \dots, x_n])$$

then we can write

$$\tau_w 1_\nu = \sum_{w' \leq w} f_{w'} \partial_{w'}$$

where $f_{w'}$ are nonzero polynomials. Since the ∂_w are linearly independent, this implies the set νB_ν is linearly independent. Thus it is a basis of $1_\nu R(\omega) 1_\nu$, and $P_\nu \alpha$ is a faithful representation of $R(\omega)$.

Example.

(a) We have $R(\omega) = k$ and $R(i) = \mathbb{Z}[X_{1,i}]$ is a polynomial ring.

(b) Let $\alpha = n_i$ for some $i \in I$. Then the only sequence in $I(\alpha)$ is $(i, i, \dots, i)$, and we have $R(\omega) \cong N H_n$.

(c) Let $\alpha = i+j$ with $i \neq j$, and choose $Q_{ij} = Q_{ji} = 1$. Then $I(\alpha) = \{(i,j), (j,i)\}$, and the algebra $R(\alpha)$ is isomorphic to

$M_2(k[x_1, x_2])$, with isomorphism given by

$$\begin{aligned} \begin{array}{|c|} \hline i \\ \hline j \\ \hline \end{array} &\mapsto \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & \begin{array}{|c|} \hline j \\ \hline i \\ \hline \end{array} &\mapsto \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ \begin{array}{|c|} \hline X \\ \hline j \\ \hline \end{array} &\mapsto \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, & \begin{array}{|c|} \hline X \\ \hline i \\ \hline \end{array} &\mapsto \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ x_1 &\mapsto x_1, & x_2 &\mapsto x_2. \end{aligned}$$

The relations are easy to check: for example, $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

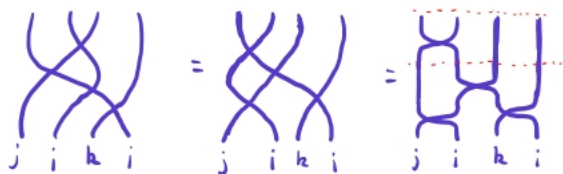
More generally, let $\alpha = i_1 + \dots + i_n$ with distinct i_r , and choose $Q_{i_r, i_s} = 1$ for $r \neq s$. Then the algebra $R(\alpha)$ is isomorphic to the matrix algebra $M_n(k[x_1, \dots, x_n])$, with the isomorphism given by

$\bar{t}_{i_1} \dots \bar{t}_{i_r} I_v \mapsto E_{s_{i_1} \dots s_{i_r}(v), v}$, where $E_{v,v}$ is the elementary matrix.

(d) Let $\alpha = \alpha_1 + \alpha_2$ be such that $Q_{ij} = 1$ for $i \in \text{Supp}(\alpha_1), j \in \text{Supp}(\alpha_2)$ (in particular $\text{Supp}(\alpha_1) \cap \text{Supp}(\alpha_2) = \emptyset$). Then $R(\alpha)$ is isomorphic to the matrix algebra of size $\binom{\text{ht}(\alpha)}{\text{ht}(\alpha_1), \text{ht}(\alpha_2)}$ with coefficients in $R(\alpha_1) \otimes_{\mathbb{K}} R(\alpha_2)$.

The isomorphism is given by $\bar{t}_{i_1} \dots \bar{t}_{i_r} I_v \mapsto E_{s_{i_1} \dots s_{i_r}(v), v}$, where $s_{i_1} \dots s_{i_r}$ is a representative of $\text{Sh}(\alpha_1) \times \text{Sh}(\alpha_2) \setminus \text{Sh}(\alpha)$

for example: let $\alpha_1 = 2i, \alpha_2 = j+k$, then



We denote by $\text{Pol}(\alpha)$ the subring of $R(\alpha)$ generated by $x_i I_v, |v| = \alpha$. Then the group S_n acts on $\text{Pol}(\alpha)$, and we can consider the fixed subring, denoted by $\text{Sym}(\alpha)$. If $\alpha = \sum_{i \in I} \alpha_i i$, then

$$\text{Sym}(\alpha) \cong \bigotimes_{i \in I} \mathbb{K}[x_1, \dots, x_{\alpha_i}]^{S_{\alpha_i}}$$

Prop. $R(\alpha)$ is a free $\text{Pol}(\alpha)$ -module of rank $n!$, with basis $\{\tau_w | w \in S_n\}$.

Proof. This follows from the basis theorem.

Prop. $Z(R(\alpha)) = \text{Sym}(\alpha)$.

Proof. Let $z \in Z(R(\alpha))$, then $z I_v = I_v z$, and $z I_v$ is central in $R(\alpha)_v$. We consider an element

$v \in I(\alpha)$ of the form $v = (i_1^{a_1}, i_2^{a_2}, \dots, i_r^{a_r})$ $a_1 + \dots + a_r = n$, $i_1, \dots, i_r$ are distinct such that $|v| = \alpha$.

In this case $R(\alpha)_v \cong \bigotimes_{t=1}^r N_{\text{ht}(\alpha_t)}$ (we can not have crossing between i_t since the target is v).

Thus $Z(R(\alpha)_v) \cong \bigotimes_{t=1}^r \mathbb{K}[x_1, \dots, x_{\alpha_t}]^{S_{\alpha_t}}$, and the composition

$$\begin{aligned} \text{Sym}(\alpha) &\hookrightarrow Z(R(\alpha)) \rightarrow Z(R(\alpha)_v) \\ z &\mapsto z I_v \end{aligned}$$

is an isomorphism.

Now by subtracting element in $\text{Sym}(\alpha)$ we can make $z I_v = 0$. For $v, v' \in I$ with $|v| = |v'| = \alpha$, if we take an element $w \in v' S_v$ with minimal length (minimal length element in a double coset), then $\tau_w I_v \in R(\alpha)_v$, and we have

$$z I_v (\tau_w I_v) = z I_v \tau_w I_v = I_v \tau_w I_v z = I_v \tau_w I_v \cdot z I_v = 0$$

Since multiplication by $\tau_w I_v$ is injective: we have $I_v \tau_w I_v \cdot I_v \tau_w I_v$ is a nonzero polynomial, we deduce that $z = 0$. This finishes the proof.

Example. Let $v = (1, 1, 2)$ and $v' = (2, 1, 1)$, then $w = \begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array}$ has minimal length in $v' S_v$,

and we take $T_{ij} = \begin{array}{c} \diagup \diagdown \\ i \quad j \end{array}$. A criterion is that no strand in w with same color intersect.

Representations. Let $Q = (I, \Omega)$ be a ^{finite} quiver and (a_{ij}) be the Cartan matrix. Let $h_{ij} = \# \{ h: i \rightarrow j \in \Omega \}$ and define

$$P_{ij}(u, v) = (v - u)^{h_{ij}}, \quad Q_{ij}(u, v) = P_{ij}(u, v) P_{ji}(v, u) = (v - u)^{h_{ij}} (u - v)^{h_{ji}} = (-1)^{h_{ij}} (u - v)^{m_{ij}}$$

where $m_{ij} = h_{ij} + h_{ji} = -a_{ij}$. We define a grading on $R(\omega)$ by

$$\deg \left(\begin{array}{c} | \\ i \end{array} \right) = 2, \quad \deg \left(\begin{array}{c} \diagup \diagdown \\ i \quad j \end{array} \right) = -a_{ij} = \begin{cases} m_{ij} & i \neq j \\ -2 & i = j \end{cases}$$

Then $R(\omega)$ become a graded algebra (one verifies that each defining relation is homogeneous).

Remark. Under this grading, if one want P_{α} become a graded $R(\omega)$ -module, the one must shift the degree on P_{α} .

We consider $R(\omega)$ -fmod and $R(\omega)$ -pmod. We will exclusively consider left graded $R(\omega)$ -modules. We write $G_0(R(\omega)) = G_0(R(\omega)\text{-fmod})$, $K_0(R(\omega)) = K_0(R(\omega)\text{-pmod})$.

Let $T: R(\omega) \rightarrow R(\omega)$ be the unique anti-automorphism determined by $T(x_i) = x_i$, $T(1_i) = 1_i$, $T(t_i) = t_i$. We can use T to define

$$\# : G_0(R(\omega)) \rightarrow G_0(R(\omega)), \quad [m] \mapsto [m^*]$$

$$\# : K_0(R(\omega)) \rightarrow K_0(R(\omega)), \quad [P] \mapsto [P^*].$$

Recall the idempotent $e_n \in N(H)$. We enlarge $I(\omega)$ by adding formal symbols: define

$$\hat{I}(\omega) = \{ v = (v_1^{(a_1)}, \dots, v_r^{(a_r)}) \mid \sum a_i v_i = \alpha \}. \quad (\text{we do not require } i_k \text{ to be distinct})$$

For $v \in \hat{I}(\omega)$, we define an idempotent

$$e_v := e_{v_1, a_1} \otimes e_{v_2, a_2} \otimes \dots \otimes e_{v_r, a_r}$$

$$e_v = \begin{array}{c} \begin{array}{|c|} \hline | \\ \hline e_{v_1, a_1} \\ \hline \end{array} \begin{array}{|c|} \hline | \\ \hline e_{v_2, a_2} \\ \hline \end{array} \dots \begin{array}{|c|} \hline | \\ \hline e_{v_r, a_r} \\ \hline \end{array} \\ \begin{array}{c} v_1 \quad v_1, v_1 \quad v_2 \quad v_2, v_2 \quad \dots \quad v_r \quad v_r, v_r \end{array} \end{array}$$

where $e_{in} \in a_i v_i R a_i v_i \cong N H a_i$. We define a module

$$P_v := (R(\omega) \cdot e_v) \left\{ \sum_{k=1}^r \frac{a_k(a_k-1)}{2} \right\} = q^{\sum_{k=1}^r \frac{a_k(a_k-1)}{2}} R(\omega) e_v.$$

We view $I(\omega) \subseteq \hat{I}(\omega)$. Then from the induction property of nil-Hecke algebra, we see that

$$P_{(v_1^{(a_1)}, \dots, v_r^{(a_r)})} = [a_1]! [a_2]! \dots [a_r]! P_{(v_1^{(a_1)}, \dots, v_r^{(a_r)})}$$

Now let $i, j \in I$ and $n = 1 - a_{ij} = 1 + m_{ij}$. We consider the following module

$$P_r := P_{(i^{(n)}, j^{(n)})} \in R(n_i + j) \text{-pmod}.$$

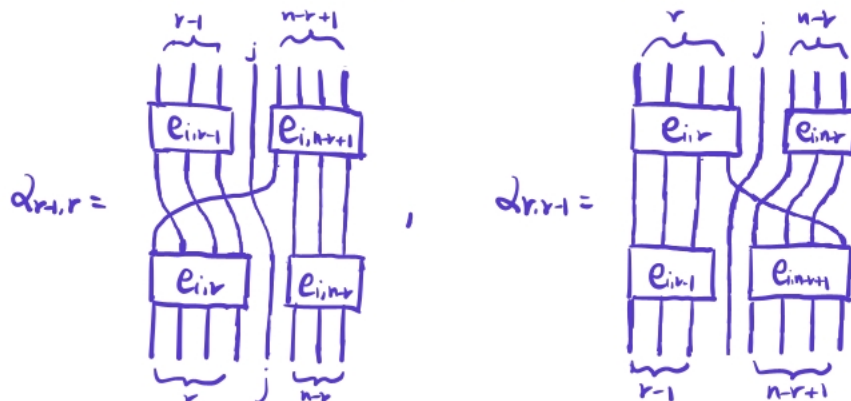
and write $e_r = e_{(i^{(n)}, j^{(n)})}$.

Prop. We have an exact sequence of projective modules

$$0 \rightarrow P_0 \xrightarrow{\alpha_{0,1}} P_1 \rightarrow \dots \rightarrow P_{r-1} \xrightarrow{\alpha_{r-1,r}} P_r \rightarrow \dots \rightarrow P_n \rightarrow 0$$

$\alpha_{1,0}$ (curved arrow from P_0 to P_1) $\alpha_{r,r-1}$ (curved arrow from P_{r-1} to P_r)

Proof. Consider $\alpha_{r-1,r} = e_{r-1} \bar{t}_1 \bar{t}_2 \dots \bar{t}_r e_r$ and $\alpha_{r,r-1} = e_r \bar{t}_1 \bar{t}_2 \dots \bar{t}_r e_{r-1}$



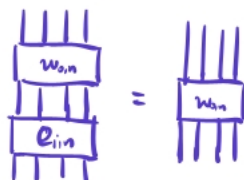
Then right multiplication gives morphisms

$$\alpha_{(r-1,r)} : P_{r-1} \rightarrow P_r, \quad \alpha_{(r,r-1)} : P_r \rightarrow P_{r-1}.$$

Lemma. The following relations are valid in NH :

$$\partial_{w_0} e_n = \partial_{w_0}$$

or diagrammatically:

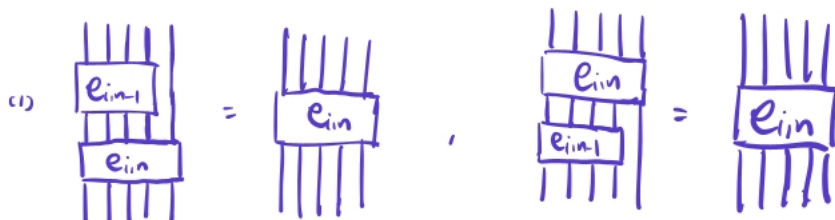


Proof. Recall that under the isomorphism $NH \xrightarrow{\sim} \text{End}_{\text{Spin}}(P_n)$, e_n corresponds to E_{w_0, w_0} . Now we have

$$\begin{aligned} \partial_{w_0}(S_w) &= \begin{cases} S_{w w_0^{-1}} & \text{if } \ell(w w_0^{-1}) = \ell(w) - \ell(w_0) \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} S_e & \text{if } w = w_0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

So ∂_{w_0} corresponds to E_{e, w_0} . Since $E_{e, w_0} \cdot E_{w_0, w_0} = E_{e, w_0}$, we obtain the claim.

Coro. We have the following diagrammatic identities:



(2)

(3)

Proof. (1)

Recall that
 $w_{i,n} = s_1 s_2 \dots s_{n-1} w_{i,n-1}$
 $= w_{i,n-1} s_{n-1} \dots s_2 s_1$

(2)

Use a different
embedding $S_{n-1} \hookrightarrow S_n$.

(3)

Return to the proposition. We have

$$\alpha_{r+1,r} = \begin{array}{c} \text{Diagram 1: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r-1}, e_{i,n+r+1}. \text{ Bottom boxes: } e_{i,r}, e_{i,n+r}. \end{array} = \begin{array}{c} \text{Diagram 2: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,n+r+1}, e_{i,n+r}. \text{ Bottom boxes: } e_{i,r-1}, e_{i,r}. \end{array} = \begin{array}{c} \text{Diagram 3: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r-1}, e_{i,n+r+1}. \end{array}$$

$$\alpha_{r,r-1} = \begin{array}{c} \text{Diagram 4: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r}, e_{i,n+r}. \text{ Bottom boxes: } e_{i,r-1}, e_{i,n+r+1}. \end{array} = \begin{array}{c} \text{Diagram 5: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r}, e_{i,r-1}. \text{ Bottom boxes: } e_{i,n+r}, e_{i,n+r+1}. \end{array} = \begin{array}{c} \text{Diagram 6: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r}, e_{i,n+r}. \end{array}$$

Note that we have

$$\alpha_{r+1,r} \cdot \alpha_{r,r-1} = \begin{array}{c} \text{Diagram 7: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r-1}, e_{i,n+r+1}. \text{ Bottom boxes: } e_{i,r}, e_{i,n+r}. \end{array} = \begin{array}{c} \text{Diagram 8: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r-1}, e_{i,n+r+1}. \end{array} = \begin{array}{c} \text{Diagram 9: } j \text{ strands on the left, } j \text{ strands on the right. Top boxes: } e_{i,r-1}, e_{i,n+r+1}. \end{array} = 0$$

Since $\begin{array}{c} \text{Diagram 10: } e_{i,n+r+1} \end{array} = \begin{array}{c} \text{Diagram 11: } \chi_{\text{Paral}}^{e_{i,n+r+1}} \\ \text{Diagram 12: } \omega_{\text{Paral}}^{e_{i,n+r+1}} \end{array} = 0$

So right multiplication by $\alpha_{r+1,r}$ gives a differential.

Now compute $\alpha_{r,r+1}\alpha_{r+1,r} - \alpha_{r,r-1}\alpha_{r-1,r}$:

$$\alpha_{r,r+1} \alpha_{r+1,r} =$$

$$\alpha_{r,r-1} \alpha_{r-1,r} =$$

So

$$\alpha_{r,r+1} \alpha_{r+1,r} - \alpha_{r,r-1} \alpha_{r-1,r} =$$

where $H_{ij}(u,v,w) = \frac{Q_{ij}(u,v) - Q_{ij}(w,v)}{u-w}$

If we write $Q_{ij}(u,v) = \sum_{a,b \geq 0} t_{ab} u^a v^b$, then

$$H_{ij}(u,v,w) = \sum_{a \geq 1, b \geq 0} t_{ab} v^b (u^{a-1} + u^{a-2} w + \dots + u w^{a-2} + w^{a-1})$$

Lemma. For $a \in \mathbb{N}$, we have

$$\begin{aligned} \begin{array}{c} \text{---} \\ | \\ \boxed{e_{i,n}} \\ | \\ \text{---} \end{array} &= \begin{cases} 0 & \text{if } a < n-1 \\ (-1)^{n-1} \begin{array}{c} \text{---} \\ | \\ \boxed{e_{i,n}} \\ | \\ \text{---} \end{array} & \text{if } a = n-1 \end{cases} \\ \begin{array}{c} \text{---} \\ | \\ \boxed{e_{i,n}} \\ | \\ \text{---} \end{array} &= \begin{cases} 0 & \text{if } a < n-1 \\ \begin{array}{c} \text{---} \\ | \\ \boxed{e_{i,n}} \\ | \\ \text{---} \end{array} & \text{if } a = n-1 \end{cases} \end{aligned}$$

Proof. Note that we have $\partial_{w,n} \partial_i = 0$ for all $1 \leq i \leq n-1$. We prove by induction on n that

$$\begin{array}{c} \text{---} \\ | \\ \boxed{w_{b,n}} \\ | \\ \text{---} \end{array} = \begin{cases} 0 & \text{if } a < n-1 \\ \begin{array}{c} \text{---} \\ | \\ \boxed{w_{b,n}} \\ | \\ \text{---} \end{array} & \text{if } a = n-1. \end{cases}$$

For $n=1$ this is clear. Now for $n>1$, we use DAH-relation to slide dots:

$$\begin{aligned}
 \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha &= \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 + \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 \\
 &= \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 + \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 \\
 &= \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 + \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1
 \end{aligned}$$

First assume $0 \leq \alpha < n-1$. Then by induction hypothesis:

$$\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n-1}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = 0$$

so we have

$$\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 + \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = 0.$$

For $\alpha=n-1$, by induction hypothesis

$$\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n-1}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n-1}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array}$$

so

$$\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 + \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha-1 = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} + \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{w_{0,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array}$$

This completes the induction step. The claim for $\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \boxed{e_{i,n}} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \alpha$ is proved similarly.

Recall that we have choose $Q_{ij}(u,v)$ to be homogeneous of degree m_{ij} . so $H_{ij}(u,v,w)$ is homogeneous of degree $m_{ij}-1=n-2$. Then from the lemma, we see that

$$\alpha_{r,r+1} \circ \alpha_{r+1,r} - \alpha_{r,r-1} \circ \alpha_{r-1,r} =$$

$$= t_{m_{ij},0}$$

$$= (-1)^{nr} t_{m_{ij},0}$$

Since $t_{m_{ij},0} \neq 0$, we get $\alpha_{r,r+1} \circ \alpha_{r+1,r} - \alpha_{r,r-1} \circ \alpha_{r-1,r} = (-1)^{nr} t_{m_{ij},0} \cdot \text{id}$, or equivalently

$$\alpha_{r,r+1} \circ (-1)^{nr} \alpha_{r+1,r} + (-1)^{nr+1} \alpha_{r,r-1} \circ \alpha_{r-1,r} = t_{m_{ij},0} \cdot \text{id}.$$

This gives a homotopy of the complex, so the complex is exact.

Coro. For $i, j \in I$ with $i \neq j$, there is an isomorphism of graded projective modules

$$\bigoplus_{r=0}^{\lfloor \frac{n}{2} \rfloor} P_{2r} \cong \bigoplus_{r=0}^{\lfloor \frac{n-1}{2} \rfloor} P_{2r+1}$$

Proof. If $m_{ij}=0$, then this means $P_{ij} \cong P_{ji}$, and the isomorphism is given by $P_0 \xrightleftharpoons[\alpha_{i,0}]{\alpha_{0,i}} P_1$

For $m_{ij} > 0$, the above computation shows the maps

$$\bigoplus_{r=0}^{\lfloor \frac{n}{2} \rfloor} P_{2r} \xrightleftharpoons[\psi]{\varphi} \bigoplus_{r=0}^{\lfloor \frac{n-1}{2} \rfloor} P_{2r+1}$$

defined by $\varphi = \sum_{r=0}^{\lfloor \frac{n-1}{2} \rfloor} \alpha_{2r,2r+1} + (-1)^{n-2r+1} \sum_{r=0}^{\lfloor \frac{n}{2} \rfloor} \alpha_{2r,2r-1}$

$$\psi = (-1)^{n-2r} \sum_{r=0}^{\lfloor \frac{n}{2} \rfloor} \alpha_{2r+1,2r} + \sum_{r=0}^{\lfloor \frac{n-1}{2} \rfloor} \alpha_{2r+1,2r+2}$$

are isomorphisms.

We now define

$$R = \bigoplus_{\alpha \in \mathbb{N}^I} R(\alpha), \quad K_0(R) = \bigoplus_{\alpha \in \mathbb{N}^I} K_0(R(\alpha))$$

For $\alpha, \beta \in \mathbb{N}^I$, we have an embedding $R(\alpha) \otimes R(\beta) \rightarrow R(\alpha+\beta)$. We denote the image of $1_\alpha \otimes 1_\beta$ in $R(\alpha+\beta)$ by $1_{\alpha,\beta}$ ($1_\alpha = \sum_{\gamma \in I(\alpha)} 1_\gamma$ is the unit in $R(\alpha)$). We then have functors

$$\text{Ind}_{\alpha,\beta}^{\alpha+\beta} = K_0(R(\alpha)) \otimes K_0(R(\beta)) \rightarrow K_0(R(\alpha+\beta)), \quad [P] \mapsto [H_{\alpha,\beta} 1_{\alpha,\beta} \otimes_{H_{\alpha,\beta}} P]$$

$$\text{Res}_{\alpha+\beta}^{\alpha,\beta}: K_0(R(\alpha+\beta)) \rightarrow K_0(R(\alpha)) \otimes K_0(R(\beta)), \quad [P] \mapsto [1_{\alpha,\beta} P].$$

Prop. The group $K_0(R)$ is a twisted bialgebra with respect to the multiplication on $K_0(R) \otimes K_0(R)$ defined by

$$(x_1 \otimes y_1)(x_2 \otimes y_2) = q^{-|y_1||x_2|} x_1 x_2 \otimes y_1 y_2$$

Proof. Use Mackey theorem.

Thm. There is an isomorphism of twisted bialgebras $\gamma: \mathcal{A}^f \rightarrow K_0(R)$ defined by $\theta_i^{(n)} \mapsto p_{i,\omega}$.

Proof. Note that for $\nu \in \hat{I}(\alpha), \nu' \in \hat{I}(\beta)$, we have

$$\text{Ind}_{\alpha,\beta}^{\alpha+\beta} p_\nu \otimes p_{\nu'} = R(\alpha+\beta) 1_{\alpha,\beta} \otimes_{(R(\alpha) \otimes R(\beta))} R(\alpha) \otimes R(\beta) e_\nu \otimes e_{\nu'} \doteq R(\alpha+\beta) e_\nu \otimes e_{\nu'} = p_{\nu\nu'}$$

so in $K_0(R)$ we have

$$[p_\nu] \cdot [p_{\nu'}] = [p_{\nu\nu'}]$$

Now the same relation follows from the complex:

$$\gamma\left(\sum_{i=0}^n \theta_i^{(i)} \theta_j^{(n-i)}\right) = \sum_{i=0}^n [p_{(i^{(n)}, j, i^{(n-i)})}] = 0.$$

To prove that γ is surjective we use a character map

$$\text{Ch}: G_0(R) \rightarrow \mathbb{Z}[\mathfrak{g}, \mathfrak{g}^*][\langle I \rangle], \quad [M] \mapsto \sum_{\nu \in \langle I \rangle} (\text{Dim } 1_\nu M) \cdot \nu$$

Here $\langle I \rangle =$ set of words on I .

Thm. The map Ch is injective.

Proof. Specialize to $q=1$ and use linearly independence of characters.

Now define a map $\gamma: \mathcal{A}^f \rightarrow \mathbb{Z}[\mathfrak{g}, \mathfrak{g}^*][\langle I \rangle]$, $\gamma(x) = \sum_{\nu \in \langle I \rangle} (\theta_\nu, x) \cdot \nu$

Then there is a commutative diagram

$$\begin{array}{ccc} G_0(R) & \xrightarrow{\gamma^*} & \mathcal{A}^f \\ \text{Ch} \downarrow & & \downarrow \gamma \\ & & \mathbb{Z}[\mathfrak{g}, \mathfrak{g}^*][\langle I \rangle] \end{array}$$

Then Ch is injective $\Rightarrow \gamma^*$ is injective $\Rightarrow \gamma$ is surjective.

There is a bilinear pair on $K_0(R) \times K_0(R)$ defined by

$$([P], [Q]) = \text{Dim } \text{Hom}_R(P^*, Q)$$

Then the extended morphism $\hat{f}: f \rightarrow K_0(R) \otimes q$ satisfies

$$(\hat{f}(x), \hat{f}(y)) = (x, y).$$

In fact, we have $(\theta_v, \theta_v) = \frac{1}{(1-q^2)^n} \sum_{w \in S_v} q^{\deg(w;v)}$, where $\deg(w;v) = - \sum_{\substack{1 \leq j < k \leq n \\ w(j) > w(k)}} v_j \cdot v_k$

and also $(p_v, p_v) = \text{Dim } 1_v R(6) 1_v = \frac{1}{(1-q^2)^n} \sum_{w \in S_v} q^{\deg(w;v)}$.

If $\hat{f}(x)=0$, then $(x,y)=0$ for all $y \in f$, hence $x=0$ by non-degeneracy. Hence γ is injective.

Thm. Assume that k has char zero. Then the canonical basis of Δf is mapped to classes of indecomposable modules in $K_0(R)$.

Example: Let $Q = \begin{smallmatrix} 1 & & 2 \\ & \nearrow & \searrow \\ & 3 & \end{smallmatrix}$, and consider the sequence $v = (1, 2, 3, 1, 2, 3)$. We have

$$\text{Dim HOM}(P_v, P_v) = (P_v, P_v) = \text{Dim } 1_v R 1_v = \frac{1}{(1-q^2)^6} \sum_{\substack{w \in S_6 \\ w(w)=v}} q^{\deg(w;v)}$$

$$\deg(w;v) = - \sum_{\substack{j < k \\ w(j) > w(k)}} v_j \cdot v_k$$

$$(14) = \begin{array}{cccc} 1 & 2 & 2 & 1 \\ & \diagdown & \diagup & \\ & & & \\ & \diagup & \diagdown & \\ & & & \end{array} \quad \text{deg}=2$$

$$(14)(25) = \begin{array}{cccccc} 1 & 2 & 3 & 1 & 2 & 3 \\ & \diagdown & \diagup & \diagdown & \diagup & \\ & & & & & \\ & \diagup & \diagdown & \diagup & \diagdown & \\ & & & & & \end{array} \quad \text{deg}=2$$

$$(25) = \begin{array}{cccc} 1 & 2 & 3 & 1 & 2 & 3 \\ & \diagdown & \diagup & \diagdown & \diagup & \\ & & & & & \\ & \diagup & \diagdown & \diagup & \diagdown & \\ & & & & & \end{array} \quad \text{deg}=2$$

$$(14)(36) = \begin{array}{cccccc} 1 & 2 & 3 & 1 & 2 & 3 \\ & \diagdown & \diagup & \diagdown & \diagup & \\ & & & & & \\ & \diagup & \diagdown & \diagup & \diagdown & \\ & & & & & \end{array} \quad \text{deg}=2$$

$$(36) = \begin{array}{cccc} 1 & 2 & 3 & 1 & 2 & 3 \\ & \diagdown & \diagup & \diagdown & \diagup & \\ & & & & & \\ & \diagup & \diagdown & \diagup & \diagdown & \\ & & & & & \end{array} \quad \text{deg}=2$$

$$(25)(36) = \begin{array}{cccccc} 1 & 2 & 3 & 1 & 2 & 3 \\ & \diagdown & \diagup & \diagdown & \diagup & \\ & & & & & \\ & \diagup & \diagdown & \diagup & \diagdown & \\ & & & & & \end{array} \quad \text{deg}=2$$

$$(14)(25)(36) = \begin{array}{cccccc} 1 & 2 & 3 & 1 & 2 & 3 \\ & \diagdown & \diagup & \diagdown & \diagup & \\ & & & & & \\ & \diagup & \diagdown & \diagup & \diagdown & \\ & & & & & \end{array} \quad \text{deg}=0$$

Thus $\text{Hom}(P_v, P_v)$ is 2-dimensional, with basis $\{v 1_v, \alpha\}$, where

$$\alpha = \begin{array}{cccc} 1 & 2 & 3 & 1 & 2 & 3 \\ & \diagdown & \diagup & \diagdown & \diagup & \\ & & & & & \\ & \diagup & \diagdown & \diagup & \diagdown & \\ & & & & & \end{array}$$

Let's see that α is an idempotent under the choice $Q_{ij}(u,v) = (-1)^{h_{ij}(u,v)} m_{ij}$: we have

$$2^2 = \text{diagram} - \text{diagram} = \text{diagram} - \text{diagram}$$

Diagrammatic equation for 2^2 involving four diagrams with three strands labeled 1, 2, 3. The first diagram is a braid. The second diagram has a red (5) label. The third diagram has a red (5) label. The fourth diagram has a red (5) label.

$$(5) = \text{diagram} - \text{diagram} - \text{diagram} + \text{diagram}$$

Diagrammatic equation for (5) involving four diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label. The third diagram has a red (5) label. The fourth diagram has a red (5) label.

$$(5) = \text{diagram} - \text{diagram} - \text{diagram} + \text{diagram}$$

Diagrammatic equation for (5) involving four diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label. The third diagram has a red (5) label. The fourth diagram has a red (5) label.

$$= \text{diagram} + \text{diagram} + \text{diagram} - \text{diagram}$$

Diagrammatic equation involving four diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label. The third diagram has a red (5) label. The fourth diagram has a red (5) label.

$$= \text{diagram} - \text{diagram}$$

Diagrammatic equation involving two diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label.

$$= \text{diagram} + \text{diagram} - \text{diagram} - \text{diagram}$$

Diagrammatic equation involving four diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label. The third diagram has a red (5) label. The fourth diagram has a red (5) label.

$$= \text{diagram} - \text{diagram}$$

Diagrammatic equation involving two diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label.

$$= \text{diagram} - \text{diagram} - \text{diagram} - \text{diagram}$$

Diagrammatic equation involving four diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label. The third diagram has a red (5) label. The fourth diagram has a red (5) label.

$$= \text{diagram} - \text{diagram}$$

Diagrammatic equation involving two diagrams with three strands labeled 1, 2, 3. The first diagram has a red (5) label. The second diagram has a red (5) label.

On the other hand, if we choose $Q_{ij}(u,v) = u+v$ for $i \neq j$, then

(b)

The diagram shows four 5-strand braid configurations, each with strands labeled 1, 2, 3, 1, 2, 3 from left to right. The first configuration has a crossing between strands 1 and 2. The second configuration has a crossing between strands 2 and 3. The third configuration has a crossing between strands 1 and 2. The fourth configuration has a crossing between strands 2 and 3. These four configurations are summed together, as indicated by the plus signs between them.

Diagram illustrating the decomposition of a braid into four terms, labeled with a red (5) and a red 1 .

$$\begin{aligned}
&= \text{Diagram 1} + \text{Diagram 2} \\
&= \text{Diagram 3} - \text{Diagram 4} + \text{Diagram 5} + \text{Diagram 6} \\
&= - \text{Diagram 7} + \text{Diagram 8} = 0.
\end{aligned}$$

So we see that $\alpha^2 = 0$, and α is not an idempotent. This means P_v is an indecomposable projective module. Yet it is not a canonical basis element, since

$$(P_v, P_v) \in \mathbb{Z} + 9\mathbb{N}[q] \neq 1 + 9\mathbb{N}[q].$$

Therefore, the choice of $\alpha_{ij}(u, v)$ is important to have the categorification of canonical basis!